

[BUG] qml.matrix does not work correctly with qml.qsvt

<https://github.com/PennyLaneAI/pennylane/issues/4206>

ROW 96

[BUG] qml.matrix does not work correctly with qml.qsvt #4206

[New issue](#)[Closed](#)

#4214



KetpuntoG opened on Jun 2, 2023

...

Expected behavior

Working with the module qml.qsvt, if I run this code, the algorithm works properly:

```
import pennylane as qml
import numpy as np
import matplotlib.pyplot as plt

def target_poly(a):
    return 0.5 * (5 * a**3 - 3 * a)

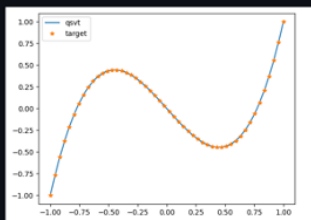
# pre-optimized angles
angles = [-1.164, 0.3836, 0.383, 0.406]

def qsvt_output(a):
    # output matrix
    out = qml.matrix(qml.qsvt)(a, angles, wires=[0], convention="Wx")
    return out[0, 0] # top-left entry

a_vals = np.linspace(-1, 1, 50)
qsvt = [np.real(qsvt_output(a)) for a in a_vals] # neglect small imaginary part
target = [target_poly(a) for a in a_vals]

plt.plot(a_vals, qsvt, label="qsvt")
plt.plot(a_vals, target, "x", label="target")

plt.legend()
plt.show()
```



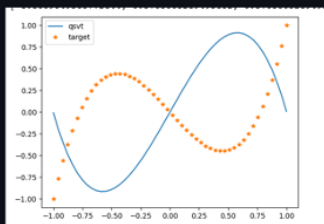
However, if you change the line :

```
out = qml.matrix(qml.qsvt)(a, angles, wires=[0], convention="Wx")
```

with

```
out = qml.matrix(qml.qsvt(a, angles, wires=[0], convention="Wx"))
```

The output is not correct:



Assignees

No one assigned

Labels

bug 🐛

Type

No type

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

[\[Bug Fix\] 'qml.qsvt\(\)' now returns a prod if 'convention='Wx''](#)
PennyLaneAI/pennylane

Notifications

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Participants



Zotero

Actual behavior

I would expect both expressions to be valid

Additional information

No response

Source code

No response

Tracebacks

No response

System information

```
Name: PennyLane
Version: 0.30.0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/XanaduAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /usr/local/lib/python3.10/dist-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, rustworkx, s
Required-by: PennyLane-Lightning

Platform info:      Linux-5.15.107+-x86_64-with-glibc2.31
Python version:     3.10.11
Numpy version:      1.22.4
Scipy version:      1.10.1
Installed devices:
```

Existing GitHub issues

- ☒ I have searched existing GitHub issues to make sure the issue does not already exist.



KetpuntoG added **bug** on Jun 2, 2023

Jaybsoni mentioned this on Jun 5, 2023

[Bug Fix] `qml.gsvt()` now returns a prod if `convention="Wx"` #4214

Jaybsoni closed this as **completed** in #4214 on Jun 6, 2023

[Bug Fix] `qml.qsvt()` now returns a prod if `convention="Wx"` #4214

<> Code

Merged Jaybsoni merged 4 commits into `master` from `fix_qsvt_matrix` on Jun 6, 2023

Conversation 5 Commits 4 Checks 0 Files changed 3

+47 -1



Jaybsoni commented on Jun 5, 2023

Contributor

Context:

`qml.qsvt()` provided functionality to change between conventions using a kwarg `convention="Wx"`. The change in convention also generated a global phase. This global phase was computed in the `qml.qsvt` function (thus it was queued) but it was NOT returned in the final operator. This meant that if a user saved the output of the function, it would be incorrect up to a global phase.

Description of the Change:

Do not silently queue the global phase, rather return an arithmetic operator which is the product of the compute qsvt operator and an identity operator with the global phase.

Related GitHub Issues:

Fixes #4206



fixed bug

8d9a87b



github-actions bot commented on Jun 5, 2023

Contributor

Hello. You may have forgotten to update the changelog!

Please edit [doc/releases/changelog-dev.md](#) with:

- A one-to-two sentence description of the change. You may include a small working example for new features.
- A link back to this PR.
- Your name (or GitHub username) in the contributors section.



Jaybsoni commented on Jun 5, 2023

Contributor Author

[sc-39532]



changelog

897f00d

Reviewers

KetpuntoG



frederikwilde



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

[\[BUG\] qml.matrix does not work correctly wi...](#)

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3 participants

